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Generative artificial intelligence in medicine

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Generative artificial intelligence (GAI) can automate a growing number of biomedical tasks, ranging from clinical decision support to design and analysis of research studies. GAI uses machine learning and transformer model architectures to generate useful text, images and sound data in response to user queries. While previous biomedical deep-learning applications have used general-purpose datasets and enormous volumes of labeled data for training, evidence now suggests that GAI models may perform better while requiring less training data—for example, using smaller, domain-specific datasets. Moreover, AI techniques have progressed from fully supervised training to less label-intensive approaches, such as weakly supervised or unsupervised fine-tuning and reinforcement learning. Recent iterations of GAI, such as agents, mixture-of-expert models and reasoning models, have further extended their capabilities to assist with complex and multistage tasks. Here, we provide an overview of recent technical advancements in GAI. We explore the potential of the latest generation of models to improve healthcare for clinicians and patients, and discuss validation approaches using specific examples to illustrate challenges and opportunities for further work.

Generative artificial intelligence (GAI) employs new types of machine-learning models to answer questions, interpret images and deliver results in the form of newly generated original text, images and sound—with remarkable quality and speed. This technology is used by hundreds of millions of users worldwide, such as for speeding up writing, answering medical questions and assisting with technical work, such as coding[1,2]. In healthcare, researchers are exploring GAI applications for many tasks, such as improving patient care and assisting with primary biomedical research. With its ability to process and generate content instantaneously, GAI could potentially reduce costs and improve the quality of healthcare processes ranging from clinical encounters and patient self-help to administrative processes, such as appointment scheduling, billing and record-keeping[1,3].

Clinical interest in GAI technology was initially piqued by the success of large language models (LLMs), such as GPT-3.5, PaLM 2 and LLaMA, which exhibited unprecedented abilities to answer challenging medical questions at the level of qualified doctors[4,5]. Subsequently, multimodal foundation models (for example, GPT-5, Gemini 2.5 Pro,

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Claude 4 and Grok 4), which can process images in addition to text, have increased the utility of GAI, including in biomedical settings[6]. Alphafold and its updated daughter models have revolutionized structural analysis of proteins and molecular interactions, paving the way for the drug discovery[7–9]. Reasoning and agentic models, such as o1 and DeepSeek-R1, exhibit enhanced ability to solve multistage problems through decomposition, iteration and use of external tools[10]. These models have achieved state-of-the-art performance in various cognitive tasks—including biomedical challenges—enabling clinicians to work together with AI teammates to boost accuracy and efficiency[10,11].

Traditionally, the management plan has been developed through a collaboration between patients and practitioners. However, a doctor–patient–AI triad could augment this process to provide optimal evidence-based, patient-centered care[12,13]. Diagnosis involves integrating patient-centric information (clinical history, laboratory results and imaging) with applicable medical knowledge (existing in up-to-date clinical articles, guidelines and textbooks), synthesized into a relevant and specific narrative, conclusion and plan. Dialog-based interfaces can maximize the utility of GAI in this context, through follow-up questions to clarify queries, reasoning and implications of conclusions. Similarly, GAI can be incorporated into biomedical scientists' workflow to accelerate discovery, hypothesis generation and reporting. Possible functions range from simple tasks, such as reformatting text, to assistance with technical tasks, such as coding and even modeling to simulate experiments and thereby maximize the efficiency of bench work[14].

In this Review, we explore recent developments in GAI, with an emphasis on new emergent abilities, as well as biomedical applications with a growing evidence base supporting their deployment and use. LLMs, foundation models and agentic systems are all discussed as examples of GAI applications in biomedical settings (see Box 1 for brief definitions of key technical terms used in the text). We specifically explore more and less successful deployments of GAI, aiming to help others learn from negative results and implementation failures. Careful, thoughtful adoption is necessary to unlock the opportunities conferred by GAI to improve the accessibility, cost, and quality of healthcare.

Technical evolution of generative artificial intelligence

Deep learning has revolutionized computational applications in medicine, particularly with respect to unstructured data, such as free text and images. Put simply, deep learning describes the data-driven tuning of relationships between virtual 'neurons,' represented in complex networks, to fulfill a defined task—such as classification of fundus photographs as normal or pathological[15]. Deep neural network architectures can represent any function: that is, any transformation of inputs into useful outputs[16]. Recently, the use of attention networks and the invention of transformers resulted in a breakthrough in natural language processing. There has since been a rapid evolution from supervised training (requiring enormous amounts of labeled data), to less label-intensive approaches using weakly and unsupervised pretraining and fine-tuning. To automate a cognitive task, AI developers design a related training task and challenge their model with that task across masses of data to improve its performance. The primary schemata for recent GAI development (Fig. 1) have involved pretraining to develop an ability to generate text, image or other data formats that are coherent; and fine-tuning (such as through reinforcement learning with expert human or AI feedback) to improve the usefulness of generated output in response to user queries[3]. Users can also use prompt engineering with deployed models to direct and optimize output to meet their needs[17].

Synthetic data systems and rule-based AI

Since 2008, there has been a growing prevalence of studies that use imputation or generate synthetic data to replace missing elements from large datasets, to facilitate analyses in the context of missing data—a common issue in clinical research[18]. A growing number of machine-learning techniques have been developed to generate synthetic data that best represent the population of interest, representing the simplest form of GAI[19]. More advanced models can generate entire datasets without including patient-identifiable data, making them suitable for development and teaching purposes[20]. Among the more commonly used architectures are variational autoencoders (VAEs) and generative adversarial networks (GANs). VAEs isolate latent variables from training data and use them to reconstruct new synthetic data[21]. This pixel-by-pixel approach often results in blurred images, limiting medical applications[22]. By contrast, GANs use a competitive strategy involving two neural networks: one generating synthetic images, and another classifying real and synthetic images. The first network is trained by the second to generate synthetic images that cannot be distinguished from real ones, enabling the production of

BOX 1

Glossary of key terms

Agentic model: an AI model capable of autonomous decision-making with limited or no human intervention required.

Attention network: an AI model that uses an 'attention mechanism' to identify more and less important parts of the input data, such as by assigning more or less weight to certain words.

Diffusion model: a GAI model that adds or 'diffuses' noise into an image and then reverses this process sequentially, thereby generating synthetic data with characteristics common to an initial training dataset.

Foundation model:an AI model trained initially on very large datasets to confer broad functionality across the modality of the training data. Subsequent fine-tuning may be undertaken to improve performance in a more specific task. Examples include LLMs.

Generative adversarial network (GAN): a machine-learning framework that pits generator and discriminator neural networks against one other to generate new synthetic data with close resemblance to an original dataset. The generator modifies input data, and the discriminator predicts whether the generated data output belongs in the original dataset.

Large language models (LLMs): text-based GAI foundation models trained and fine-tuned to provide useful responses to user queries.

Neural network: the architectural basis of modern artificial intelligence, with computationally represented nodes (artificial 'neurons'), usually arranged in layers, that have tunable relationships between one another to transform data for useful purposes.

Reasoning model: a subset of foundation models that are fine-tuned to solve multi-step reasoning tasks, such as by enforcing chain-of-thought narration in model processing or output.

Retrieval augmented generation: the technique of mandating reference to a specified information source (such as clinical practice guidelines) to improve the accuracy and relevance of GAI output.

Transformers model (transformers): a neural-network-based architecture, which is the technical basis for most widely disseminated foundation models, that allows for entraining of sequential construction of useful data on the basis of lexical tokens or other component parts of larger data elements.

Variational autoencoder (VAE): a subset of artificial neural network architectures which maps information into a latent space before reconstructing information into similar but different information, generating new synthetic data.

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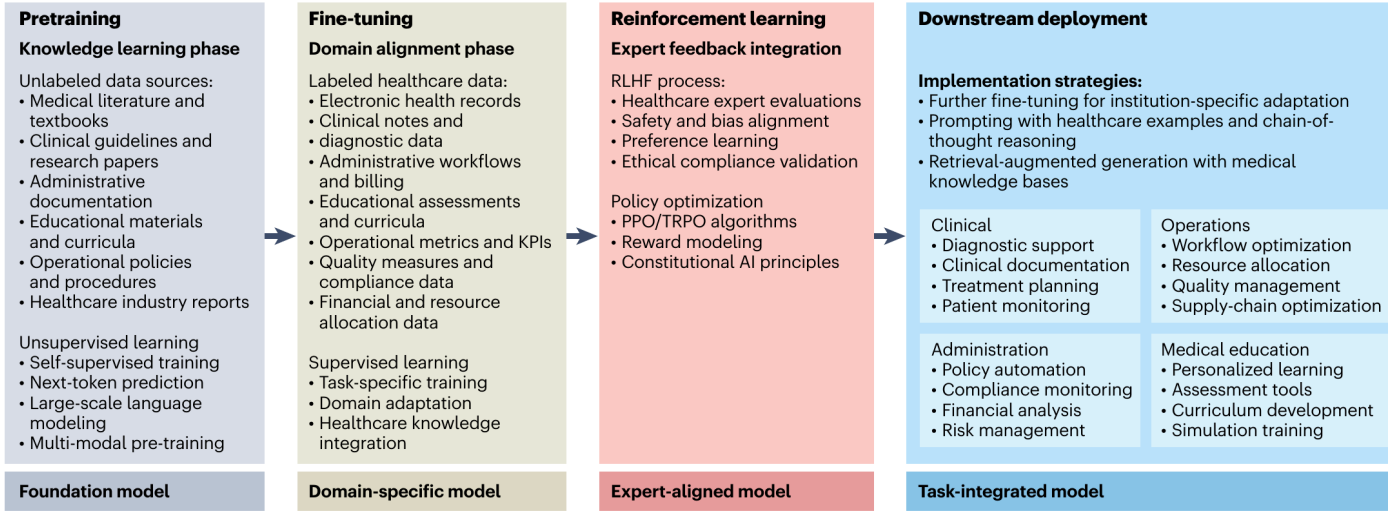

Fig. 1 | Overview of the GAI development pipeline. The figure shows key steps from initial foundation model development to their deployment in specialized healthcare applications across clinical care, operations, administration and medical education. KPIs, key performance indicators; PPO, proximal policy optimization; TRPO, trust region policy optimization.

highly detailed, realistic images[23]. However, statistical 'noise' leads to inconsistent fidelity of generated images, and there is a risk of reproducing patient-identifiable features from images used during training[24].

Diffusion models have recently emerged as state-of-the-art architectures for generating images that closely resemble real examples (such as of radiographs or computed tomography images). These models work by sequentially adding noise to and subtracting it from an image, generating variation through loss and stochastic replacement of information[25]. This two-step procedure results in better-quality images with a broader variety than those generated by GANs or VAEs[25,26]. Although its computation tends to be slower than that of lighter-weight architectures, it can still be run locally. Commonly used diffusion-model applications, including Stable Diffusion 3 and DALL-E 3, perform poorly when tasked with biomedical imaging; specific training of bespoke models is needed to use these types of model to generate realistic synthetic medical imaging[27,28]. With this training, diffusion models can generate synthetic images with realistic anatomical details—even for three-dimensional modalities, such as computed tomography and magnetic resonance imaging, providing valuable data for training diagnostic algorithms[27,29].

Many rule-based GAI applications are already used for clinical purposes that involve natural language processing. These rule-based bots prioritize safety over flexibility, making them particularly effective in formulaic or algorithmic contexts, as well as in situations with established techniques for steering conversations (such as cognitive behavioral therapy). Indeed, more than 10,000 mental-health applications collectively have millions of users who often pay subscription fees Relatively few of these applications undergo formal clinical validation, but examples of trialed platforms include https://www.wysa.com/ and https://woebothealth.com/ (ref. 30). Another successful example is Dora, an automated telephone-call system for cataract surgery follow-up. Dora uses a predetermined set of conversational elements and management options to identify patients in need of clinical follow-up in multiple hospitals in the United Kingdom[31]. Although emerging foundation models might have enhanced flexibility and broader capabilities, some developers are actively delaying the replacement of existing rules-based systems until there are better safety assurances[32].

Foundation models with growing capabilities

Foundation models now represent the frontier of GAI. In general, foundation models exhibit large transformer-based architectures and are trained on large datasets of one or more modalities, developing abilities to produce new but coherent information in these same modalities[5]. The weakly supervised or unsupervised pretraining and fine-tuning processes that underpin foundation models distinguish them from previous machine-learning architectures. The earliest iteration of foundation model that gained widespread attention and use was the LLM, which was the initial technical underpinning for chatbot applications such as ChatGPT and Google Bard. LLMs provide an instructive example of development paradigms which apply to foundation models more generally[3] (Fig. 2).

Pretraining involves tasking an LLM with a word-related task across voluminous text-based datasets. Tasks require the model to predict missing words or portions of words ('tokens') in human-written material[33,34]. Datasets are generated by extracting text from internet-based and private resources, including clinical practice guidelines, peer-reviewed journal articles and medical textbooks, as well as non-medical text. Subsequent fine-tuning aims to promote generation of useful output in response to user queries. Fine-tuning may use illustrative input–output pairs produced by humans, or automate this process through reinforcement learning from human feedback (RLHF)[35]. In RLHF, discrete evaluation models are trained using data from humans who score a limited set of outputs in response to inputs. These models can then replicate human-like scoring to assess and fine-tune LLM responses. Furthermore, human involvement in RLHF can itself be automated, in reinforcement learning from AI feedback (RLAIF)[36]. Conversely, state-of-the-art mixture-of-expert models (for example, DeepSeek-R1) dispense with the critic model required for RLHF or RLAIF, in favor of a group relative policy optimization (GPRO) process—in which multiple outputs are directly compared with one another to encourage production of outputs with favored characteristics, such as accuracy and relevance[37]. This promotes selective recruitment of portions of the model architecture on the basis of user queries to efficiently provide an optimal response[10]. All these fine-tuning processes may be tailored depending on the desired characteristics of the model, such as factuality, relevance and tone.

Similar processes can be applied to develop vision–language models, audio-language models and other multimodal foundation models (Fig. 2). For instance, foundation models have been pretrained on various formats of clinical imaging (with word-based tokens replaced by other forms of information) and can be fine-tuned to perform classification tasks with performance comparable to that of state-of-the-art conventional deep-learning methods. An early example is RETFound, which was trained in a 'fill in the blank'

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Pretraining	Fine-tuning	Reinforcement learning	Deployment
Text Data sources: • Medical literature textbooks • Clinical guidelines protocols • Medical research databases • Drug information systems Techniques: • Autoregressive language modeling • Masked language modeling (BERT) • Next-sentence prediction	**Text fine-tuning** Clinical data: • Electronic health records • Clinical notes narratives • Discharge summaries • Lab-result annotations Techniques: • Named-entity recognition • Clinical-text classification • Medical-question answering	**Text RLHF** Human feedback: • Clinical accuracy assessment • Safety and terminology correctness • Medical-bias evaluation • Ethical-compliance review Techniques: • Proximal policy optimization • Clinical preference learning • Constitutional AI alignment	**Text deployment** Applications: • Clinical documentation AI • Medical chatbot virtual assistants • Literature search synthesis • Clinical decision support Deployment techniques: • A/B-tested learning • RAG with medical knowledge bases • Prompt engineering optimization
Image Data sources: • Medical imaging datasets • X-rays, CT scans • Pathology-slide images • Histopathology slides Techniques: • Vision transformer (ViT) • Masked autoencoders (MAE) • Contrastive learning	**Image fine-tuning** Clinical data: • Annotated X-rays, radiographs • Labeled MRI CT scans • Pathology slide annotations • Dermatology image datasets Techniques: • Medical-image classification • Semantic segmentation • Object-detection localization	**Image RLHF** Human feedback: • Radiologist evaluations • Diagnostic accuracy assessment • Clinical relevance scoring • Visual attention validation Techniques: • Vision-based PPO • Human preference ranking • Visual attention alignment	**Image deployment** Applications: • Radiology AI assistants • Pathology analysis systems • Dermatology screening tools • Medical image generation Deployment techniques: • Zero-shot classification • Visual prompting • Few-shot adaptation
Video Data sources: • Surgical procedure videos • Diagnostic microscopy • Medical training footage • Patient consultation videos Techniques: • Video transformers (ViViT) • Three-dimensional convolutional networks • Temporal modeling	**Video fine-tuning** Clinical data: • Annotated surgical videos • Labeled endoscopy clips • Motion-tracking clinical data • Gait analysis datasets Techniques: • Action recognition • Temporal segmentation • Motion pattern analysis	**Video RLHF** Human feedback: • Surgeon technique assessment • Temporal accuracy evaluation • Patient safety scoring • Clinical workflow validation Techniques: • Video-based PPO • Temporal reward modeling • Sequence preference learning	**Video deployment** Applications: • Surgical guidance systems • Endoscopy AI assistants • Patient-monitoring tools • Rehabilitation monitoring Deployment techniques: • Real-time video analysis • Temporal prompting • Motion-pattern recognition
Audio Data sources: • Heart sounds murmurs • Respiratory recordings • Patient interview recordings • Voice biomarker datasets Techniques: • Audio transformers • Mel-frequency analysis • Spectral masking	**Audio fine-tuning** Clinical data: • Labeled heart murmurs • Annotated lung sounds • Speech biomarker annotations • Clinical audio transcripts Techniques: • Audio classification • Signal filtering • Acoustic feature learning	**Audio RLHF** Human feedback: • Clinician audio assessment • Diagnostic quality scoring • Pattern-recognition accuracy • Clinical relevance scoring Techniques: • Audio-based PPO • Signal-quality modeling • Acoustic preference learning	**Audio deployment** Applications: • Cardiac monitoring AI • Respiratory analysis systems • Voice-based diagnostics • Clinical transcription AI Deployment techniques: • Audio prompting • Signal preprocessing • Few-shot audio learning
Multimodal Data sources: • Text–image pairs • Video–audio datasets • Multi-modal clinical records • Cross-modal annotations Techniques: • CLIP-style learning • Cross-modal attention • Unified transformers	**Multimodal fine-tuning** Clinical data: • Radiology reports and images • Clinical notes and audio • Surgery videos and text • Patient records and biomarkers Techniques: • Vision–language tasks • Multi-modal fusion • Joint representation learning	**Multimodal RLHF** Human feedback: • Cross-modal consistency • Integrated clinical reasoning • Holistic patient assessment • Multi-modal safety scoring Techniques: • Multi-modal PPO • Cross-modal alignment • Joint preference optimization	**Multimodal deployment** Applications: • Comprehensive clinical AI • Integrated diagnostic systems • Multi-modal medical assistants • Holistic patient monitoring Deployment techniques: • Cross-modal prompting • Fusion optimization • Integrated decision support

Fig. 2 | GAI development pipeline based on specific modalities. The key steps in development pipeline of GAI models include: (1) pretraining with careful selection of data sources; (2) fine-tuning with clinical data and context-specific information; (3) reinforcement learning, which relies on human input (required) to evaluate for aspects such as accuracy, relevance and bias; and (4) deployment, which are crucial steps for clinical translation. CT, computed tomography; MRI, magnetic resonance imaging; CLIP, contrastive language–image pretraining; RAG, retrieval-augmented generation.

image-modeling task in which the model was exposed to fundus photographs with missing portions and tasked with reconstructing the missing pixels[38,39]. Other foundation models have been developed to work with computed tomography, optical coherence tomography, pathology slides, ultrasounds and X-ray images[40–45]. Many proprietary models—including those used to drive popular chatbots—are trained and fine-tuned with multimodal data, allowing interoperability and diversification of tasks that applications can assist with[46–48]. This allows users to input speech and images in addition to text, and also expands the range of application outputs.

Early anecdotal evidence and more recent formal studies of LLMs have revealed that they perform better at many cognitive tasks in which prompts mandated 'chain of thought' reasoning (explicitly processing problems and solutions in a logical, step-by-step manner)[49]. Researchers have since incorporated chain-of-thought reasoning into fine-tuning to promote this behavior, improving reasoning ability as

well as capabilities of leveraging external tools to generate solutions[10]. Reasoning models such as DeepSeek-R1, Gemini 2.5 Pro, GPT-5, Claude 4 and Grok 4 are trending examples of 'agentic' AI, which require less user feedback and can autonomously solve problems and complete tasks[50]. Agentic models can query search engines to retrieve relevant information, implement code in a virtual workspace to trial solutions or even leverage automated machine learning to build AI models specifically for a given task[51,52]. In the field of medicine, there is hope that agentic AI will work collaboratively with clinicians, patients and scientists to tackle complex problems and promote innovation[11,53–56].

Model distillation for clinical tasks

The ongoing development of models with growing reasoning and response abilities has generated optimism that 'generalist' medical AI—essentially medical foundation models that can automate diverse medical tasks with little or no specific training—will begin to be deployed in clinical contexts[57]. However, because close oversight is needed to safeguard patients from potential harm caused by autonomous systems, GAI is likely to be initially implemented in small, siloed functions with carefully and narrowly defined boundaries. Therefore, a more efficient and practical solution for healthcare settings could rely on smaller models that are developed specifically to optimize performance in a highly specific medical task[58].

Smaller models with comparable performance to that of industrial flagship foundation models can be engineered relatively simply through a process called model distillation, whereby a small, open-source language model is fine-tuned on a set of outputs generated by flagship models[59,60]. Domain-specific fine-tuning can facilitate superior performance in clinical tasks relative to state-of-the-art models[61]. Such fine-tuning is typically limited by lack of access to patient data owing to data-privacy governance, although ongoing efforts aim to broaden access to large, multimodal clinical datasets[62].

The potential benefits of smaller GAI models are manifold. Smaller models are less computationally expensive than are industrial LLMs or other foundation models; therefore, lower associated costs could broaden access, particularly in lower-income settings[63]. In addition, smaller models can be deployed locally in air-gapped systems in clinical organizations, minimizing security risk and privacy concerns associated with uploading data online[64]. A modular approach using small models for well-defined functions could also facilitate troubleshooting without compromising broader systems, because component models can be interrogated individually (in contrast to relying on a single large model with broader functionality). However, local deployment entails costs and requires infrastructure that might not be accessible, potentially leading to a need to rationalize expenditures by reducing other clinical investments[65].

The technical limitations of smaller foundation models can be overcome in part by users applying validated techniques during prompting. One limitation is that smaller models tend to have a lower context length—meaning they have a stricter limit on the amount of text that can be inputted or processed at one time. Users can utilize chunking strategies, processing information in smaller segments, to overcome this limitation[66]. Smaller models also tend to produce less-desirable outputs in terms of responding appropriately and flexibly to queries, as well as raw recall of accurate specialist knowledge[67]. Prompt-engineering strategies, such as encouraging chain of thought, negative bounding to inhibit undesirable behavior and retrieval augmented generation, can mitigate these issues[17,68]. Specific education of clinicians and patients can be undertaken to teach these techniques, to help ensure that tools maximize their potential[69,70].

Clinical applications of generative artificial intelligence

GAI applications are yet to be accepted and used widely in autonomous clinical roles, but are used widely for administrative tasks, and by patients and practitioners for medical conversations through chatbots (rather than internet search engines)[71–73]. Most validation studies of GAI evaluate a narrow subset of potential roles (such as clinical decision-making or documentation), and although there are many examples of GAI exceeding clinician performance in individual tasks, this is not grounds for replacing clinicians in their complex, holistic roles[12]. Moreover, small retrospective studies are liable to bias and overfitting that limits generalizability, and model performance in studies may not translate into real-world settings[74]. Nonetheless, GAI's assistive role in healthcare is growing, and considering existing applications and barriers to deployment can help inform research and development of more useful systems.

Clinical support

Medical GAI garnered initial interest after LLM chatbots achieved passing level marks in examinations taken by medical students and doctors[75,76]. Since then, developers have undertaken specific training and fine-tuning to improve GAI performance in these examinations; the latest models are now approaching or exceeding the performance of expert clinicians[4,77,78]. Although examination performance is a poor surrogate for actual clinical ability, one study directly compared a GAI model and clinicians in responding to patient enquiries posted on a social-media forum—and found that the model provided higher-quality and more-empathetic responses than clinicians did (assessed in a blinded fashion by healthcare professionals)[79]. Since then, a growing number of studies have evaluated GAI's potential for providing clinical advice in different contexts. Although these models offer greater scalability than do human clinicians, many of the studies are poorly conducted (lacking standardized evaluation processes) and reported (inaccessible models and absent description of prompt engineering), and offer little useful information to guide implementation and subsequent development[80].

Early results from a prospective study illustrate the strengths and weaknesses of GAI in providing clinical advice and guidance[81]. For example, clinicians and AI, challenged with virtual-reality cardiopulmonary resuscitation scenarios, performed best when clinicians oversaw AI that provided management guidance; this scenario was superior to clinicians working alone or autonomous AI[82]. Similarly, an economic analysis of clinical AI in specific contexts, such as diabetic retinopathy screening, suggests that AI–human collaboration is superior to either working alone[83]. However, LLMs tasked with making challenging diagnoses based on a documented history, examination and laboratory results did not improve physicians' performance, indicating that GAI could be less useful in situations lacking specific algorithms to guide reasoning[84]. Experiments with radiologists also suggest that clinicians undervalue and separate AI predictions from their own reasoning, limiting the benefits of AI predictions even where these predictions are highly accurate[85]. When the diagnostic reasoning of LLMs is specifically interrogated, deficiencies relative to experienced clinicians are revealed even where LLMs reach the correct answer, illustrating an important gap requiring further development and validation work[86]. The advent of reasoning models—which are specifically trained to better mimic logical thought processes recognizable by humans—has improved performance in complicated cognitive tasks, such as clinical reasoning; further improvement might be possible by teaching clinicians how best to prompt models to optimize responses[10,69,87].

GAI clinical functions outside question-answering and provision of advice are relatively understudied[80,88]. However, researchers are applying foundation models to tasks that could improve healthcare quality. Foresight is a predictive clinical transformer trained with electronic health records (EHRs) to forecast future medical events, procedures and diagnoses with high accuracy[61]. Foresight 2 exhibits superior performance over an industrial foundation model (GPT-4), highlighting the value of using domain-specific data with smaller models, rather than relying on flagship proprietary platforms[89]. However,

Foresight's development has been halted owing to concerns regarding unauthorized data use—highlighting the ongoing deliberation and negotiation of stakeholders to navigate preservation of data privacy while promoting innovation.

Other, better-studied GAI applications concern text-based chatbots, which are widely used in mental-health counseling and surgical follow-up[30,31]. These can be used with or without clinician administration, empowering patients to take charge of their care and obtain prompt access to psychological interventions[90]. Foundation models offer opportunities to develop chatbot platforms with greater capabilities and flexibility[91,92]. However, substantial risks merit careful validation and monitoring. For instance, a report of a chatbot user committing suicide after being encouraged by GAI has highlighted significant concerns about the potential consequences of automated mental-health counseling[93]. A safer deployment plan could use GAI as an advisory tool for counselors or therapists, potentially increasing their efficiency and capacity to consult patients while retaining human oversight of dialogue[94,95].

Medical education

Currently, clinicians in training learn through self-directed study and supported training with lectures, small-group tutorials and simulated or real patients. GAI can assist with all of these scenarios, leveraging its indefatigability and flexibility with regards to tone and level of discourse[91]. Medical students given feedback from GAI chatbots exhibited superior performance to their peers who were working on the same training sessions but did not receive GAI feedback. Differences emerged after just four sessions—highlighting the potential of foundation models to improve the provision of tailored clinical education[96].

A recent rapid review of the literature base indicated that more papers opined on potential use-cases, rather than reporting experimental tests of GAI in educational contexts[97]. Studies most commonly appraise GAI for personalized tutoring or as a medical search engine, for content development for educators and for simulation of patient interactions to facilitate low-stakes communication practice[97]. GAI 'tutors' for anatomy education and case-based teaching have been developed, although there is limited robust validation to justify deployment for medical students or doctors in training[98,99]. Important risks include hallucination and propagation of inaccurate, harmful information; this problem is more common when models are required to recall specific facts, such as supporting references[100]. In addition, to minimize any risk of compromising medical education, proving that students benefit from GAI is essential before mandating or endorsing its use.

Administrative assistance

Clinicians are plagued by growing administrative responsibilities, including documentation, billing, coding, scheduling and inventory management. Administrative burdens impact healthcare professionals by reducing job satisfaction and increasing the likelihood of errors that might affect patient care[101]. GAI can streamline these tasks and thereby improve how clinicians use their time. Because many of these tasks do not directly affect clinical care, it can be argued that validation requirements for GAI deployment in these settings should be lower[81]. However, the dramatically increased administrative burden that came with EHR deployment in healthcare demonstrates the critical importance of evidence-based deployment to ensure that workflow interventions improve clinicians' experience at work[102].

GAI excels at processing and producing text at superhuman scale and speed and might therefore help alleviate the documentation burden in healthcare. Potential applications range from on-demand chart review and note generation, to automation of EHR functions, such as generation of medical histories and clinical coding[103]. Studies of ambient GAI scribes—that process speech during consultations to produce draft documentation—suggest that clinicians highly approve of this use of technology, owing to work and time-savings, good quality of documentation and empowerment to be more present with patients[104,105]. GAI exhibits remarkable summarization ability, with one study demonstrating superiority to clinicians in terms of quality and efficiency[106]. In general, GAI appears to produce highly readable documentation that contains the most-important points that clinicians wish to emphasize, which has been tested in discharge summaries and informed-consent notes[107,108].

Clinical coding is a labor-intensive administrative task and is crucial for recordkeeping, public health, research and billing[109]. Because codes must conform exactly to dictionaries, such as the International Classification of Diseases 10, hallucination or other failures lead to unacceptable performance. Proprietary LLMs, including GPT-3.5, GPT-4, Gemini Pro and Llama 2, exhibit match rates lower than 50%, likely owing to the tokenization process during training—in which text is split into small units around the same size as words or clinical codes, but without preserving the intrinsic structure of the coding system[109,110]. To enhance performance, specific training and fine-tuning of symbolic foundation models that process clinical codes as discrete units separate from natural language, is essenetial[111]. Downstream benefits of improved coding models could extend to other processes, such as audit, insurance claims, cost calculation and research, all of which depend on faithful documentation of diagnoses and intervention.

Three important risks must be considered with deploying GAI for administrative clinical tasks, even in instances in which performance seems superior to clinical experts. First, performance is liable to degradation in non-English languages, largely because most pretraining and fine-tuning data are in English[3,108]. In addition, because LLMs struggle with ambiguity—in which source text is non-specific—as well as hallucinations or invented facts, delegation to GAI entails a risk of generating and promulgating false information. Mitigating strategies could include a human-in-the-loop, who has clinical oversight and responsibility; having another or the same GAI system verify outputs 'in parallel'; or leveraging chains of GAI 'in series' to improve text quality[112]. Finally, owing to the idiosyncratic formatting and storage structures in EHRs, performance validated in 'ideal' test settings with reproduced data might not reflect real-world settings, particularly with different EHR platforms[113]. Ideally, models should be trained, fine-tuned and validated specifically in EHR platforms—which is challenging owing to information governance policies and the need for access to sufficient computing resources—to ensure that models can work effectively with patient data.

Primary research

GAI is accelerating biomedical research by automating key components, such as hypothesis generation, study design, data analysis and report writing. Various proof-of-concept implementations demonstrate GAI's potential in research: appraising and designing new machine-learning architectures, linking with a robotic system to fully automate theorizing and proving structure–function relationships of proteins, and even designing therapeutics that could treat disease[14,114,115]. With the availability of automated machine learning, GAI systems might be able to autonomously construct deep-learning models for an unlimited variety of tasks[52,116]. GAI agents could thereby function as virtual research collaborators, broadening access to multidisciplinary expertise by taking advantage of their general training, which spans across all fields of academic study[117]. Not all of the impact of this automation will be positive: a dramatic increase in formulaic reports of studies analyzing publicly available datasets has been observed since the proliferation of GAI chatbots, with many of these studies being of poor quality and likely originating from paper mills and citation farms[118].

Synthetic data produced by GAI could facilitate more-ambitious studies than are currently feasible. For instance, synthetic data might augment, or even replace, sensitive datasets derived from patient records, permitting research that can inform clinical practice—such as randomized control trials, which frequently struggle to enroll a

sufficient number of participants—or aid development of new interventions, such as computational systems that require data to train or validate[20]. However, there are potential problems with relying on synthetic data, which is, by definition, not collected from real patients. Synthetic data might not contain the full range of idiosyncratic differences between individuals, and the performance of models that are trained exclusively on synthetic data tends to degrade with more training[119]. Because synthetic data are frequently derived from real text, imaging and other information from patients, they can contain patient-specific features and thereby release confidential information that could be identifiable[120].

GAI has also formed the technical basis for new research tools that have permitted unprecedented research in molecular biology. AlphaFold and its daughter models accurately predict protein structures and can now model protein–protein interactions on-demand; these investigations previously required extensive laboratory experimentation[7,8,121]. ESM3 is a multimodal GAI model that reasons over protein sequences, structures and functions. ESM3 demonstrates abilities to engineer new proteins with similar functions to existing species and can be customized by users providing free-text prompts. ESM3 has been used to generate new fluorescent proteins whose structures are significantly different from those of any existing species, indicative of genuine creation rather than imitation[122]. Evo and Evo 2 are genomic foundation models that leverage training on 300 billion nucleotides to generate and analyze DNA sequences at the whole-genome scale. Evo can thereby design and predict the efficacy of gene-editing systems such as CRISPR–Cas9, enhancing the potential of genetic engineering to lead to new medical therapies[123,124]. Use of data gathered from large numbers of experiments—many of which do not lead to published results—could conceivably lead to proliferation of foundation models that can augment laboratory and clinical research.

Finally, GAI can assist methodological research, literature review and report writing by accelerating literature searches, abstract screening and narrative syntheses of published results. LLMs exhibit comparable performance in identifying papers relevant to a review question when compared with authors of Cochrane Library systematic reviews who have domain-specific expertise[112]. Various research models offer synthesis functionality to provide a preliminary overview of any field of inquiry, and comparative results suggest that these overviews are of comparable quality to summaries produced by humans, such as on Wikipedia articles[125]. Ongoing work will integrate these abilities to develop agentic models that can generate useful hypotheses and design and simulate methods to answer important scientific questions[126,127]. An early multi-agent 'AI co-scientist' built around Gemini 2.0 has demonstrated the ability to identify new pharmacological targets, and even to design new drugs with promising in vitro activity, suggesting that GAI can accelerate biomedical discovery and development of new therapeutics[128].

Evaluation and quality assurance

Establishing a robust evaluation framework that encompasses technical, clinical, regulatory and ethical aspects is essential for ensuring that GAI interventions are safe, effective and reliable, with appropriate return on investment to justify integration into existing or new workflows. A step-wise approach, analogous to the process of clinical training with increasing responsibility, provides an instructive framework[129]. Evaluation of clinical applications will likely need to go beyond mere 'task-based certification' to encompass comprehensive frameworks that assess real-world clinical impact[130].

Preclinical evaluation (research and development phase)

Standardized testing and artificial but instructive clinical scenarios may be used to prove that an application can provide useful assistance, and that its functionality is not compromised at predictable 'pain points'. Most published studies involving GAI currently fall into these categories, with few studies involving real patient data, and even fewer being prospective clinical studies[80,131].

For quantitative evaluation, conventional statistical measures, including accuracy, sensitivity, specificity, area under the receiver operating characteristic curve), precision, recall and F1 score, may be used for amenable tasks[132]. However, while task-specific algorithms can still be evaluated with conventional metrics, these methods frequently fail to capture the performance of foundation models. Qualitative assessment may be required to provide a more holistic assessment of GAI applications (see Table 1 for examples of qualitative and quantitative metrics)[78,79]. These metrics could also be grouped as intrinsic metrics, extrinsic metrics and emerging metrics specific to multimodal clinical foundation models.

Intrinsic metrics use principles borrowed from the field of linguistics to measure coherence and meaningfulness of output[133]. These methods may provide a statistical score based on overlapping words (for example, BLEU (Bilingual Evaluation Understudy), ROUGE (Recall-Oriented Understudy for Gisting Evaluation) or METEOR (Metric for Evaluation of Translation with Explicit Ordering)), the frequency of characters that should be replaced to optimize coherency (for example, Levenshtein distance) or sentence structure (for example, CIDEr (Consensus-based Image Description Evaluation))[134–139]. However, the objectivity and reliability of these algorithmic scoring systems comes at the expense of specificity to context and task.

Conversely, extrinsic metrics incorporate the context of the task and stakeholder perspectives to provide a more insightful score, generally at the expense of increased subjectivity and indeterminate scoring[133]. For instance, expert human raters could be tasked with assessing GAI output with reference to one or more desired characteristics, as exemplified by the SCORE (safety, consensus, objectivity, reproducibility, explainability) framework[140] (Table 1).

Alternatively, LLMs can themselves be used to apply extrinsic metrics, either by automating calculation of linguistic metrics (for example, BERT-SCORE[141]), or through more-sophisticated analysis of adherence to defined ground truths (for example, systematic reviews, clinical practice guidelines, reputable primary sources) with logical consistency and relevance to the subject at hand[142–147]. There is growing interest in this role of 'LLM as a judge,' which offers a cost-effective, consistent and scalable approach to evaluation of complex task performance[148]. A recent validation of LLM-as-a-judge for evaluation of GAI-generated summaries of EHRs exhibited strong inter-rater reliability compared with expert human evaluators, even in cases that required advanced clinical reasoning and domain-specific expertise[149]. Further work is necessary to enable interpretability of automatic extrinsic metrics, as well as to develop validation benchmarks to justify their use.

With the emergence of multimodal foundation models, there is a demand for updated metrics to facilitate evaluation of clinical applications. For a comprehensive overview of foundational metrics for clinical GAI assessment, Abbasian et al. have provided a summary grouped under the headings of accuracy, trustworthiness, empathy and performance[150]. A multi-metric approach to evaluation is very likely necessary to overcome the limitations of any single system[151]. This can allow researchers to highlight strengths and weaknesses of a new application with greater specificity, helping to guide subsequent development work and anticipate issues with clinical deployment.

Clinical evaluation and implementation

Once an application has demonstrated good performance in test settings and there is a clear plan for implementation, clinical validation is necessary. Initially, close oversight is recommended, particularly for systems that influence clinical decision-making[129]. For clinical interventions that impact diagnosis, investigation or treatment, randomized clinical trials, which permit objective assessment of the effectiveness and safety of a new system, will be likely necessary to justify deployment[81]. Many previous trials of AI-based interventions were relatively

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Table 1 | Quantitative and qualitative evaluation metrics for GAI

Metrics[a]	Purpose	Units
Quantitative evaluation		
AUROC (area under the receiving operating curve)	To evaluate the model's ability to discriminate classes across different thresholds.	0 to 1
AUPRC (area under the precision-recall curve)	To evaluate the model's ability to discriminate the positive (usually minority) class.	0 to 1
Precision (positive predictive value)	The proportion of model-identified elements that are relevant.	0 to 1
Sensitivity (recall)	The proportion of true positive elements that are correctly identified by the model.	0 to 1
Specificity	The models' ability to correctly identify elements without a condition (true negatives).	0 to 1
F1 score	A metric combining precision and recall.	0 to 1
Dice coefficient	Also known as the Dice similarity coefficient, this statistical metric is used to measure the similarity between two sets.	0 to 1
BLEU	Evaluates machine translation quality by measuring *n*-gram precision: how many *n*-grams (sequences of words) in the AI-generated text appear in the reference text.	0 to 1
ROUGE	Designed for text summarization. Measures overlap between AI-generated text and reference text using recall.	0 to 1
METEOR	Evaluates machine translation quality incorporating linguistic features and placing more emphasis on recall.	0 to 1
BERT-SCORE	Computes a similarity score between AI-generated and reference text using contextual embeddings (semantic equivalence).	0 to 1
Qualitative evaluation		
Safety[b]	Evaluate the degree of hallucination.	Likert Scale 1–5 1, Strongly disagree 2, Disagree 3, Neutral 4, Agree 5, Strongly agree
Consensus and context[b]	Response is aligned with clinical evidence, professional consensus and context.	
Objectivity[b]	Response is objective and unbiased against any condition, device or demographic.	
Reproducibility[b]	Contextual consistency of responses after repeated generation to the same question.	
Explainability[b]	Justification of response, including reasoning process and additional supplemental information.	

This list includes metrics that are frequently used in existing studies, but it is not exhaustive. [a]Linguistic metrics are not strictly distinct from one other and can cover overlapping aspects of model evaluation [b]There is currently no gold-standard evaluation method for these metrics.

small (often single-center), used non-clinical endpoints and provided limited information on demographics—making it difficult to evaluate generalizability[152]. Larger studies with clinical primary endpoints (for example, mortality or morbidity) and transparent reporting would represent the most convincing evidence supporting deployment of GAI. Once validated robustly, autonomous deployment with less direct oversight can be planned, with structured revalidation and surveillance for potential adverse consequences, analogous to longitudinal stage 4 clinical trials[129]. To improve the standard of study design, conduct and reporting, many reporting guidelines—some specific to GAI—have been developed through expert consensus-seeking exercises, such as those published by the EQUATOR Network[153–156]. In addition, the development of a multicentric benchmarking framework (MedHELM) by researchers at Stanford University allows researchers to evaluate their models on a broad range of real-world tasks[157].

For non-clinical interventions aimed at improving clinicians productivity' or quality of working life, it could be argued that randomized trials are not necessary[81]. For instance, models that draft correspondence while clinicians retain responsibility and oversight can be evaluated using extrinsic metrics[79]. However, prospective randomization is the most definitive way to analyze causal relationships related to a new intervention, and comparable A/B testing has been well established in adjacent fields[158,159]. These types of study are important before deploying GAI systems at scale, because even well-intentioned technological 'solutions' can inadvertently lead to problems such as inefficiency, degraded quality of documentation and clinician burnout[160].

Thorough evaluation of concerns about bias and fairness is essential for clinical GAI applications, to avoid inequitable benefits and potential harm to patients, for example, due to algorithmic bias as a result of under-representation of marginalized groups, or inequitable access to beneficial interventions owing to socioeconomic factors or variable mistrust in GAI among different communities. A growing number of initiatives to promote active consideration and action to remedy these inequities are available to support clinicians, researchers and policymakers, including STANDING-TOGETHER, FUTURE-AI, CARE-AI and SCORE[161–163]. Through standardization of high-quality work in these domains, there is hope that the field as a whole will advance in addressing problems concerning bias and fairness. A promising approach is the creation of shared benchmarking datasets that test performance in specific clinical tasks.

In addition to quantitative and qualitative assessments of GAI model performance, it is important to evaluate application safety in terms of the risks that deployment entails. These risks vary with the type of model (closed versus open source), data input (and related consent or de-identification procedures) and plan for ongoing monitoring to exclude performance drift. Finally, health economic analysis is an essential precursor to deployment—particularly in view of the substantial resource requirement for many GAI systems[164]. Many GAI systems require justification of considerable upfront investment for information technology, manpower, governance and ongoing updates. Understanding the cost of implementation and relating this to other potential uses of resources ensures that decisions are rationalized on the basis of what benefits patients most. Considering the anticipated return of investment in direct and indirect domains is important—particularly for interventions that substantially change workflows or patient outcomes.

Future opportunities

Although GAI has revolutionized many industries, including finance, education, retail, transportation and technology, uptake in medicine has been relatively slow[165]. This is likely in part owing to the difficulty in engineering models with sufficient performance to match that of clinicians in a complicated and frequently ambiguous field, which also depends on the trust of patients and practitioners, without leading to adverse or inequitable outcomes. Research and development efforts should be directed in four broad areas to translate technology into useful clinical applications.

First, although much attention has been placed on model development, subsequent deployment in real-world settings is relatively understudied[80,152,166]. Robust clinical validation in pragmatic trials and ongoing monitoring—to mitigate any performance degradation and unintended consequences of deployment—will be essential[81]. Second,

opaque and unclear reporting is a widespread concern. To maximize transparency, methodology and datasets used in GAI model development should ideally be made available, detailing which models were used, how they were customized and what infrastructure was used to deploy them. This will allow researchers to replicate results and build on other teams' work[155,167]. Third, improving AI literacy will enable clinicians and patients to make the best use of GAI tools, but this requires targeted efforts from medical schools and throughout clinical training[69]. Finally, comprehensive and coherent governance structures are required to allow developers to invest in GAI development and deployment without fears regarding future permissibility. The European Union Artificial Intelligence Act provides an early example, requiring providers of high-risk AI systems to report serious incidents to active market surveillance authorities[168].

GAI technology continues to evolve with new advancements, such as large concept models, allowing for superior reasoning and contextual understanding[169], and agentic GAI with greater autonomy[170]. Further work is necessary to develop GAI applications that integrate into existing clinical workflows, address ethical and privacy concerns, as well as to agree a system of governance that preserves incentive structures for researchers and developers while ensuring that patients remain safe and clinicians benefit from evidence-based changes to their work patterns.

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